MIL-C-24231/21D(SH)
15 May 1987
SUPERSEDING
MIL-C-24231/21C(SHIPS)
30 August 1976
(See note 4)

MILITARY SPECIFICATION SHEET

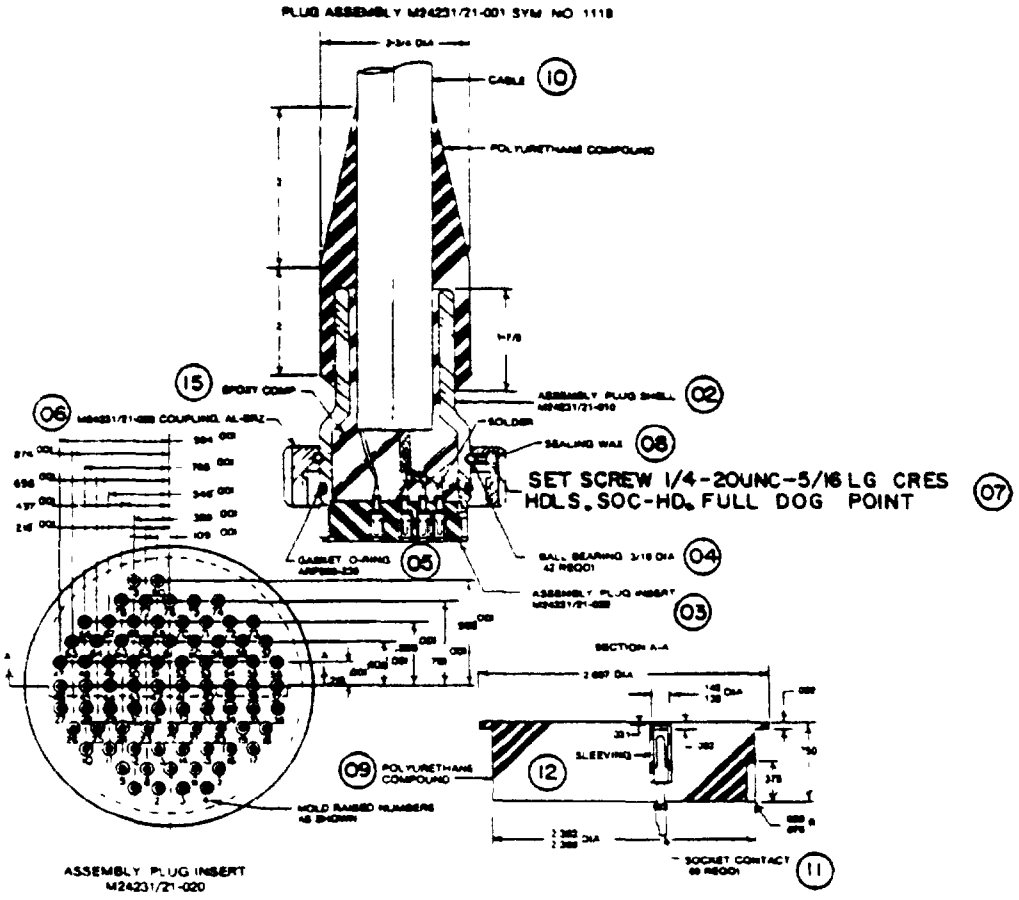
CONNECTOR, PLUG, TYPE I, MOLDED, 80-CONDUCTOR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the plug described herein shall consist of this specification and the latest issue of MIL-C-24231.

AMSC N/A FSC 5935
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SH 9688

FIGURE 1. Connector, plug and insert assembly, molded.

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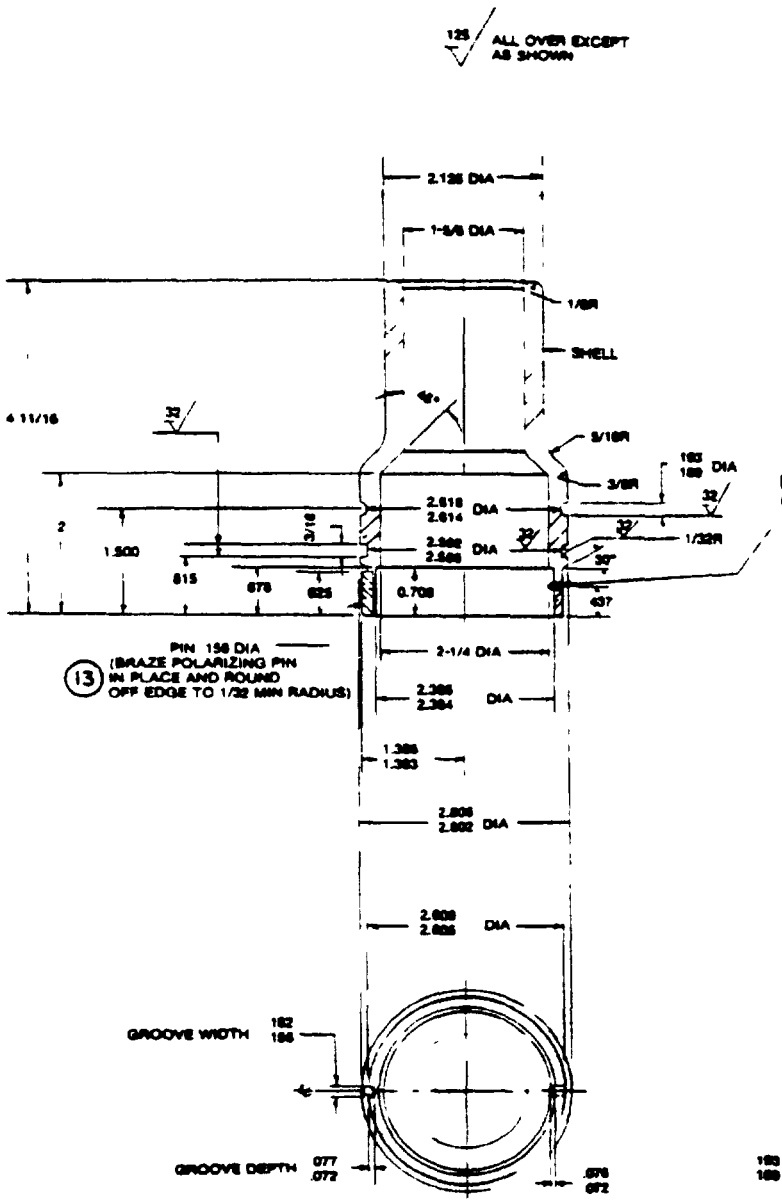


FIGURE 2. Assembly plug shell - M24231/21-010. 02

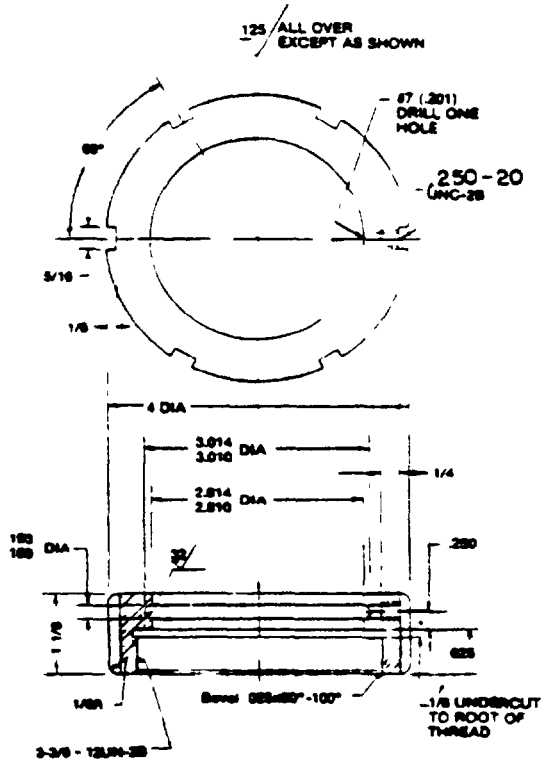


FIGURE 3. Coupling - M24231/21-030. 06

SH 9688-1

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REQUIREMENTS:

1. Dimensions and configuration: See figures.
2. If the plug shell is machined in the unaged condition it shall be heat-treated to a hardness within the range of 250 to 320 Brinell or 24 to 34 Rockwell C measured in the ball bearing groove area.
3. Each coupling shall be examined for cracks in accordance with the liquid penetrant method of MIL-STD-271. Finding of cracks shall be cause for rejection.
4. Hydrostatic pressure test specified in MIL-C-24231 for mated plug and receptacle assembly (first article inspection) and for type I molded plugs (quality conformance inspection) shall be reduced to 1000 lbf/in^2 for 24 hours and 1 hour, respectively.

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TABLE I. Parts list.

Item 1/ no.	Description 2/	Quantity for M24231/21- 3/ 001	Material/specification	Remarks
01	Plug, 80-conductor	1	-----	See figure 1
02	Shell, plug	1	Ni-Cu-Al, QQ-N-286	See figure 2
03	Insert assembly, plug	1	See items 09, 11, and 12	See figure 1
04	Bearing, ball 3/16 diameter	42	Bronze, MIL-B-1083	See figure 1
05	Gasket, O-ring	1	MS28775-230	Lubricate with silicone compound, MIL-S-8660
06	Coupling	1	Ni-Al-BRZ, QQ-C-465	See figure 3
07	Screw, set, 1/4-20UNC by 5/16 Lg, HDLS-Soc-Hd, full dog point	1	CRES, coml	See figure 1
08	Wax, sealing	As required	Coml	See figure 1
09	Compound, polyurethane	As required	Type I, category A, MIL-M-24041	See figure 1
10	Cable, type ECM	As required	MIL-C-915/54	See figure 1
11	Contact, socket	80	MIL-C-24231	See figure 1, section A-A

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/21- 3/ 001	Material/specification	Remarks
12	Sleeving, socket no. 9, 0.118 id by 9/16 Lg	As required	Type F, subform U_a, grade a, class II, category 1, clear, MIL-I-631	See figure 1, section A-A
13	Pin, polarizing 0.156 diameter	1	Ni-Cu, class A, QQ-N-281	See figure 2
14	Pin, polarizing 0.156 diameter, length to suit	1	Ni-Cu, class A, QQ-N-281	See figure 2
15	Compound, epoxy	As required	Cast epoxy resin compound 4/	See figure 1

1/ Item numbers are the same as encircled numbers shown on figures.
2/ See applicable figure or table listed in remarks column for part numbers and dimensions.
3/ Quantity is for one complete assembly of each size.
4/ Arcoset No. 200, M.M. Reif and Co., Rockledge, PA, or Phillicast No. 2, Philadelphia Resin Corp., Montgomeryville, PA, or equal.

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NOTES:

1. All dimensions are in inches.
2. Unless otherwise indicated, tolerances shall be as follows:

   (a) Decimals - plus or minus 0.005 inch.
   (b) Fractions - plus or minus 1/64 inch.
   (c) Angular dimensions - plus or minus 1/2 degree.

3. These connectors are for maintenance purposes only.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5935-N279-21)

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